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CO1 - ASSIGNMENT

Senario- Smart Access Control for a smart campus University

TASK-1:- Mathematical Formulation of Formal languages and Automata

Introduction

Automata theory deals with mathematical models used to recognize patterns in string. In a smart access control system, finite automata are used to validate binary access code containing 0s and 1s

Input Alphabet and string

An alphabet is a finite set of symbols

$\Sigma = \{0, 1\}$

A string is a finite sequence of symbol from an alphabet

example : 0, 1, 00, 01, 101, 1101

Formal language

A language is a set of string formed using an alphabet

$L = \{w \in \{0,1\}^* \mid w \text{ starts and ends with same symbol}$

4. DIFFERENCE BETWEEN String and Language

Finite string	Empty string	Language
Sequence of symbol	No symbols	Set of strings
101	ε	
	$\|\varepsilon\|=0$	

5. Five - Tuple Representation

A finite automaton is represented by

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where $Q \rightarrow$ Set of states

$\Sigma \rightarrow$ Input alphabets

$\delta \rightarrow$ Transition function

$q_0 \rightarrow$ Initial state

$F \rightarrow$ final state.

Role of States

States represents the current conditions

Transition define movement between the states

Initial state is starting point

final state indicates acceptance.

Sample String

String	Result
0	Accepted
1	Accepted
01	Rejected
101	Accepted
110	Rejected

TASK-2 Mathematical design and construction of DFA

DEFINITION

DFA is a finite automata in which exactly one transition exists for every state and input symbol.

Language :- Binary string that starts and end with same symbol.

States

q_0 - Initial state
q_1 - Start and end with 0
q_2 - Start with 0 end with 1
q_3 - Start and end with 1
q_4 - Start and 1, end with 0

Five tuple :-

$$M = (Q, \Sigma, \delta, q_0, F)$$

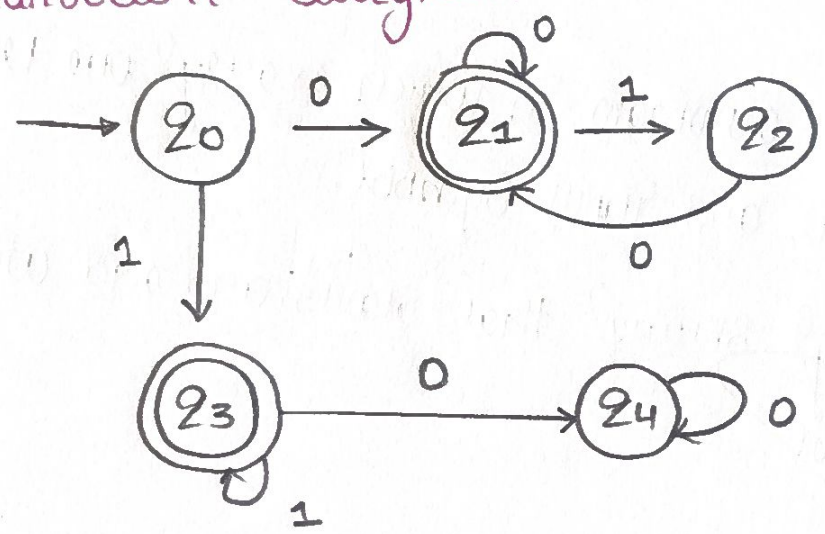
$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\} \quad F = \{q_1, q_3\}$$

Transition Table

State	0	1
→ q_0	q_1	q_3
* q_1	q_1	q_2
q_2	q_1	q_2
* q_3	q_4	q_3
q_4	q_4	q_3

Transition diagram



Input Tracing

101

$q_0 \rightarrow q_3 \rightarrow q_4 \rightarrow q_3$

String accepted

011

$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_2$

String rejected

The DFA accepts strings whose first and last are same symbol

Task 3 - Non-deterministic finite automata for pattern recognition

Introduction:-

A non-deterministic finite automaton (NFA) is a finite automaton that can have multiple possible transitions for same input symbol

The objective is to recognize binary strings containing the substring 101

Identification of states

four states are required:

q_0: Initial state Searches for the patterns

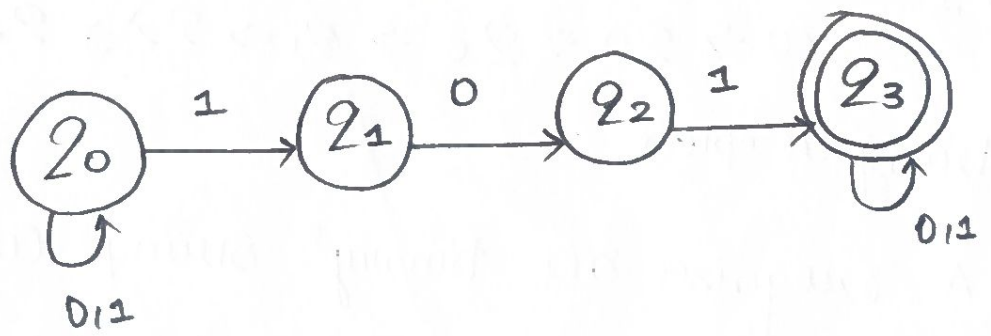
q_1 - recognizes 1

q_2 - recognizes 10

q_3 - recognizes 101 (final state)

$F = \{q_3\}$

NFA Transition diagram



Transition Table

State	0	1
→ q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$
* q_3	$\{q_3\}$	$\{q_3\}$

Mathematical Representations

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\}$$

$$F = \{q_3\}$$

Input traces

Input = 1101

$$q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$$

String accepted

11101

$$q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$$

String accepted

The NFA recognizes all binary strings containing the substring 101 using multiple possible computation paths

Task 4 - Conversion of NFA into DFA

Definition

Subset construction converts an NFA into an equivalent DFA by combining NFA states into subsets

Reachable Subsets

A = {q0}

B = {q0, q1}

C = {q0, q2}

D = {q0, q1, q3}

E = {q0, q2, q3}

F = {q0, q3}

DFA Transition Table

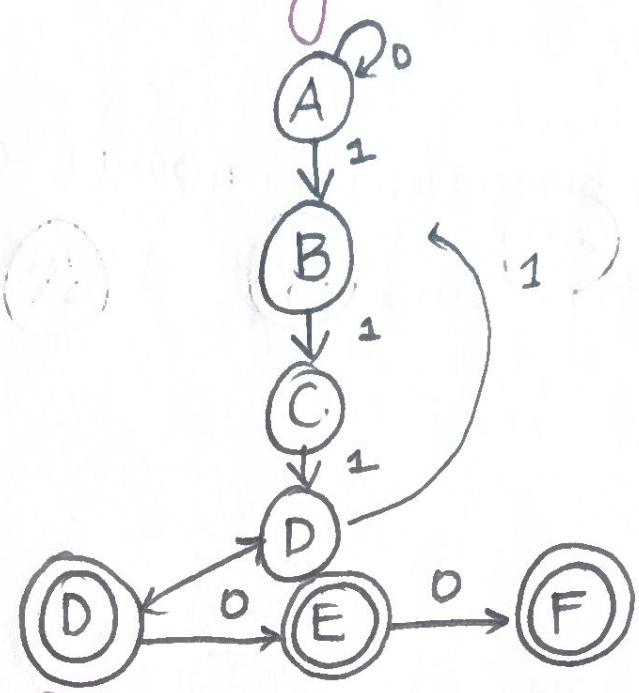
State	0	1
→ A	A	B
B	C	B
C	A	D
*D	E	D
*E	F	D
*F	F	D

Initial State and final State

Initial State : A

final States : D E F

Transition diagram



Example Verification

1101

A → B → B → C → D

String accepted

1001

A → B → C → C → B

String rejected

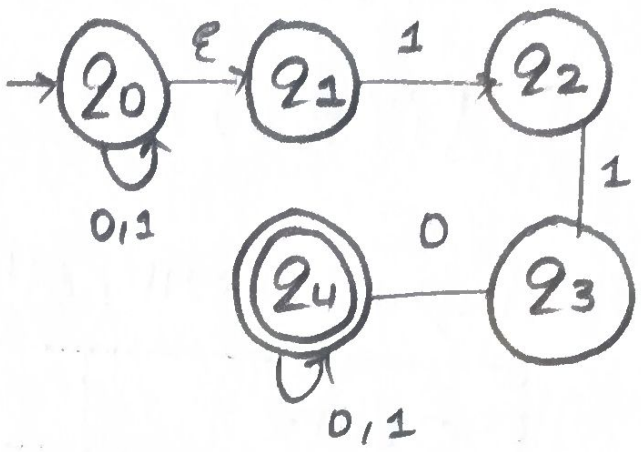
Thus the converted DFA is equivalent to the orgina NFA and accepts strings containing 101.

Task 5 - ε NFA to Equivalent NFA

An ε-NFA allows transitions without consoming an input symbol. ε-closure helps remove these transitions

Required Substring

110



States

q_0 → Initial state

q_1 → first 1

q_2 → Second 1

q_3 → Recognizes 11

q_4 → Recognizes 110

ε-NFA transition Table

State	0	1	ε
→ q_0	$\{q_0\}$	$\{q_0\}$	$\{q_1\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$	$\emptyset$
q_3	$\{q_4\}$	$\emptyset$	$\emptyset$
* q_4	$\{q_4\}$	$\{q_4\}$	$\emptyset$

ε-closure (q_0) = $\{q_0, q_1\}$

ε-closure (q_1) = $\{q_1\}$

ε-closure (q_2) = $\{q_2\}$

ε-closure (q_3) = $\{q_3\}$

ε-closure (q_4) = $\{q_4\}$

Conversion Formula

$\delta'(q,a) = \varepsilon\text{-closure}(\text{move}(\varepsilon\text{-closure}(q),a))$

Equivalent NFA transition table

State	0	1
$\rightarrow q_0$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2, q_4\}$
q_1	$\emptyset$	$\{q_2, q_4\}$
q_2	$\emptyset$	$\{q_3, q_4\}$
q_3	$\{q_4\}$	$\emptyset$
$*q_4$	$\{q_4\}$	$\{q_4\}$

$F' = \{q_4\}$

The converted NFA has no ε-transitions

String	ε-NFA	NFA
110	Accepted	Accepted
1010	Rejected	Rejected
0110	Accepted	Accepted

The equivalent NFA accepts the same language Without using ε-transitions.

48/50